UNIVERSITI PUTRA MALAYSIA

"BERILMU BERBAKTI"

CCS3700 – EMBEDDED SYSTEMS PROGRAMMING

LAB 6

TEMPERATURE SENSOR

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SUBMITTED DATE:

19/5/2026

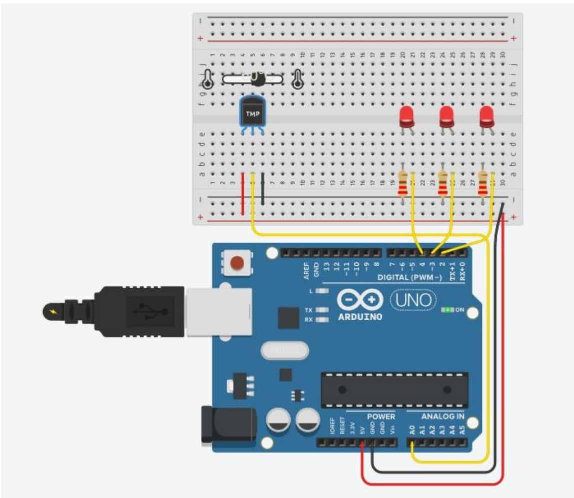
WHAT IS NEEDED IN THIS LAB

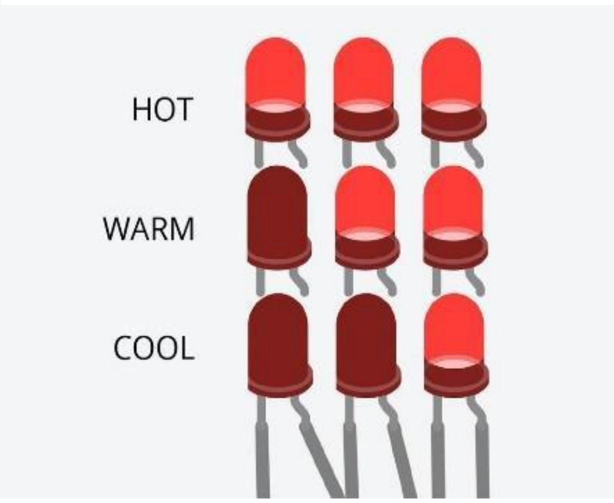
COMPONENTS	FUNCTIONS
TMP36 TEMPERATURE SENSOR	To check temperature and return the output
3x LED	To show the output
3x RESISTORS	To give a bit of resistance for the 5V
1x ARDUINO UNO	Main Controller
1x BREADBOARD	To distribute the powers
POSITIVE RAIL	Give 5V to LED and temp sensor
NEGATIVE RAIL	Give GND to LED and temp sensor

WHICH PIN GOES TO WHERE

COMPONENTS	PINS
TMP36 TEMPERATURE SENSOR	POS RAIL, A0, NEG RAIL
RESISTOR 1, 2 & 3	NEG RAIL, LED CATHODE SHORT LEG
LEFT LED	D4, RES 1
MIDDLE LED	D3, RES 2
RIGHT LED	D2, RES 3

SAMPLE CIRCUIT AND NOTES





Set up the hardware as shown above. Write a program and compile to the Arduino indicating the temperature change (as seen with the simulated temperature sensor (TMP36) above using the given chart below. The temperature change will be indicated through the 3 LEDs as shown below

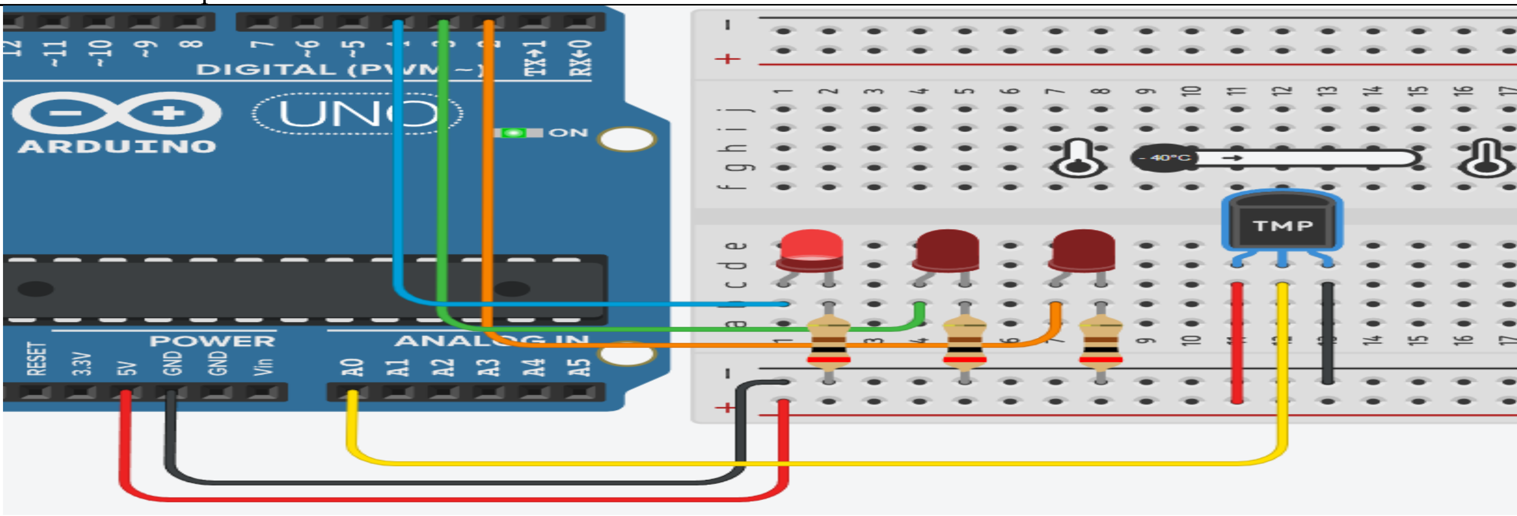
Hot = 60 C and above
Warm = 50 C to 59 C
Cool = 49 C and below

Start the simulation and ensure there is output to the serial monitor.
Notes:

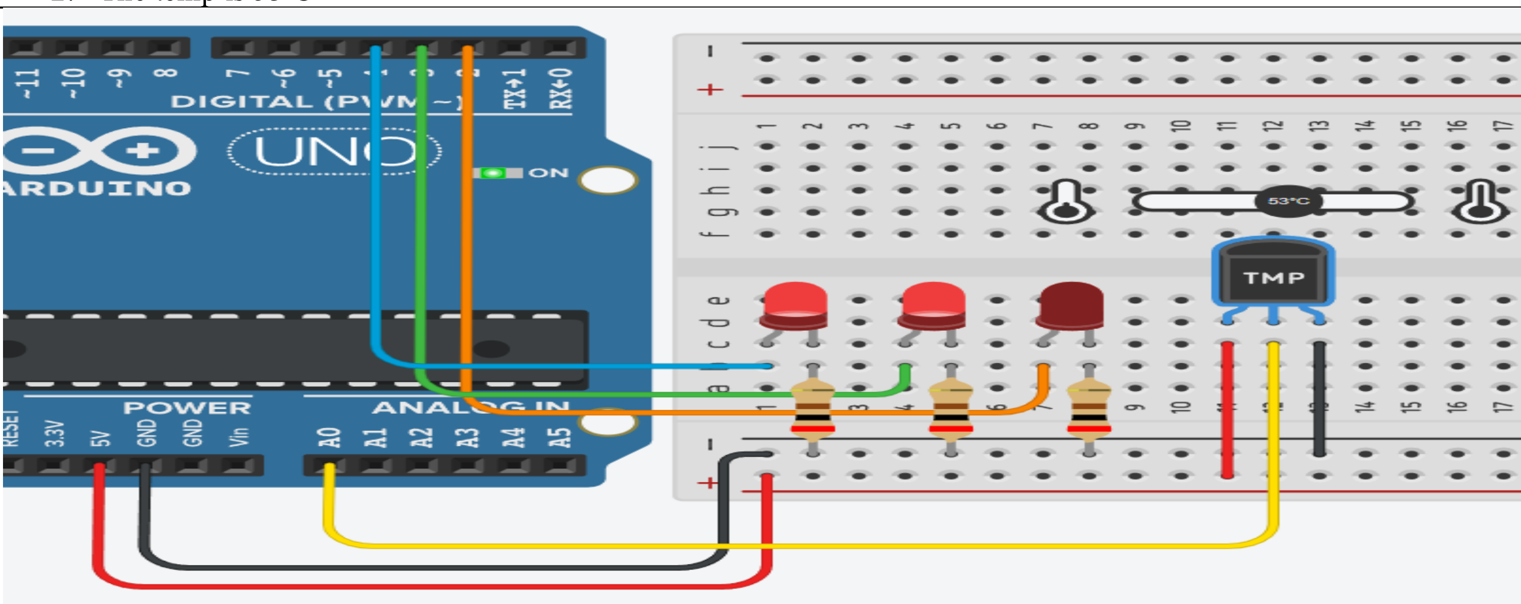
1. The formula for converting between Celsius and Fahrenheit is F = (C * 9) / 5 + 32.
2. Printing to the serial monitor helps you observe the temperature change more granularly than the LED states show alone.
3. Click on the TMP36 sensor to enable the variable temp pop-up bar on TinkerCAD.

OUTPUT

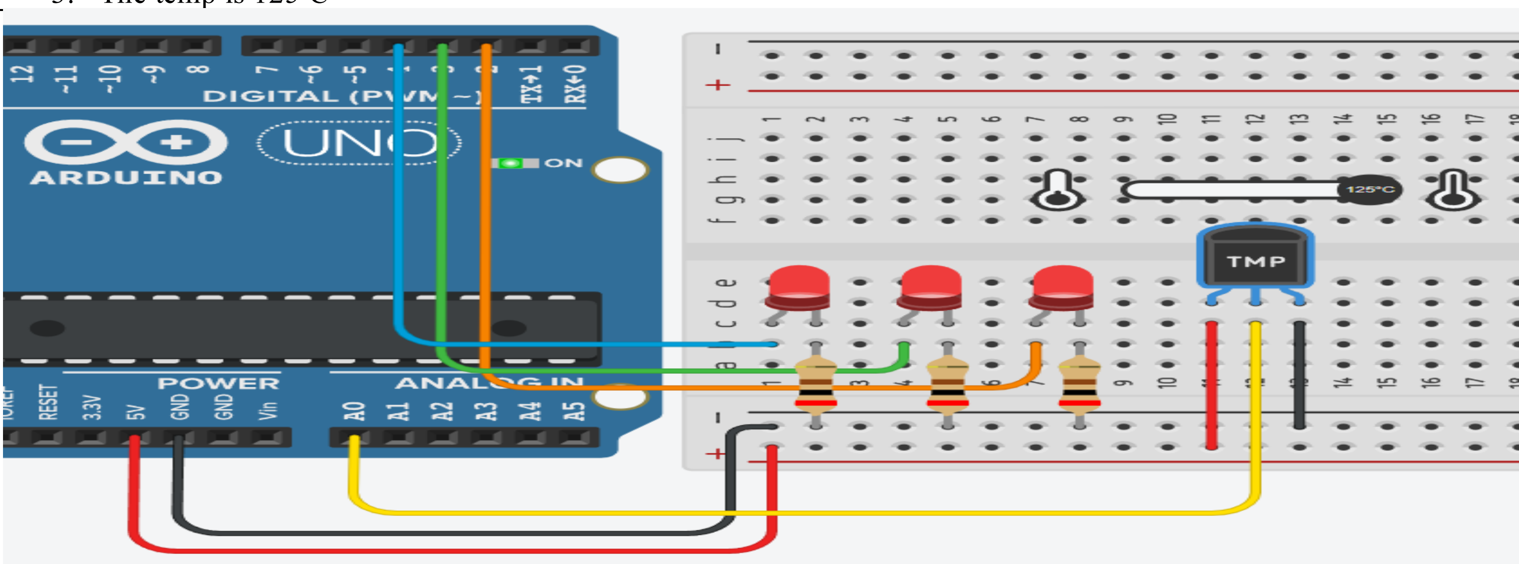
1. The temp is -40 C



2. The temp is 53 C



3. The temp is 125 C



THE WHOLE CODE

```
// Define the pins
const int sensorPin = A0;
const int ledCool = 4; // Left LED
const int ledWarm = 3; // Middle LED
const int ledHot = 2;  // Right LED

void setup() {
  // Start the serial connection to the computer
  Serial.begin(9600);

  // Set the LED pins as outputs
  pinMode(ledCool, OUTPUT);
  pinMode(ledWarm, OUTPUT);
  pinMode(ledHot, OUTPUT);
}

void loop() {
  // 1. Read the analog value from the sensor (0-1023)
  int sensorVal = analogRead(sensorPin);

  // 2. Convert the reading to voltage
  float voltage = (sensorVal / 1024.0) * 5.0;

  // 3. Convert the voltage to temperature in Celsius
  // The TMP36 has a 500mV offset and 10mV/degree scale
  float temperatureC = (voltage - 0.5) * 100;

// 4. Convert Celsius to Fahrenheit
  float temperatureF = (temperatureC * 9.0) / 5.0 + 32.0;

// 5. Print the results to the Serial Monitor
// Formatting to match the lab document: "25 C, 77 F"
Serial.print(temperatureC, 0); // Print without decimals
Serial.print(" C, ");
Serial.print(temperatureF, 0);
Serial.println(" F");

// 6. Control the LEDs based on the temperature
if (temperatureC >= 60.0) {
  // HOT: 60 C and above -> Turn on all 3 LEDs
  digitalWrite(ledCool, HIGH);
  digitalWrite(ledWarm, HIGH);
  digitalWrite(ledHot, HIGH);
}
else if (temperatureC >= 50.0) {
  // WARM: 50 C to 59 C -> Turn on 2 LEDs
  digitalWrite(ledCool, HIGH);
  digitalWrite(ledWarm, HIGH);
  digitalWrite(ledHot, LOW);
}
else {
  // COOL: 49 C and below -> Turn on 1 LED
  digitalWrite(ledCool, HIGH);
  digitalWrite(ledWarm, LOW);
  digitalWrite(ledHot, LOW);
}

// Wait a second before the next reading
delay(1000); }
```

SERIAL MONITOR

COLD	MODERATE	HOT
-40 C, -40 F	52 C, 126 F	84 C, 183 F
-33 C, -27 F	52 C, 126 F	70 C, 158 F
-17 C, 2 F	58 C, 136 F	66 C, 150 F
1 C, 33 F	58 C, 136 F	69 C, 156 F
19 C, 66 F	57 C, 134 F	80 C, 176 F
31 C, 88 F	56 C, 133 F	87 C, 188 F
37 C, 98 F	56 C, 133 F	96 C, 205 F
40 C, 104 F	56 C, 133 F	101 C, 214 F
40 C, 104 F	56 C, 133 F	103 C, 217 F
	56 C, 133 F	107 C, 224 F
	56 C, 133 F	113 C, 235 F
	56 C, 133 F	125 C, 257 F
	56 C, 133 F	125 C, 257 F
		125 C, 257 F

ENTIRE CODE: https://github.com/aslamrofi/ESP-LAB/tree/LAB6

END OF LAB 6